

## Gases

1. A weather balloon is initially filled with 10.0 L helium gas at 1.00 atm and 20.0 °C. The balloon rises to an altitude where the pressure is 26.0 torr and temperature is -50.0 °C. What is the volume (in L) of the balloon at the higher altitude?

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{(1.00 \text{ atm}) \cdot (10.0 \text{ L})}{(20.0 + 273.15) \text{ K}} = \frac{\left(26.0 \text{ torr} \times \frac{1 \text{ atm}}{760 \text{ torr}}\right) \cdot V_2}{(-50.0 + 273.15) \text{ K}}$$

$$\frac{(1.00 \text{ atm}) \cdot (10.0 \text{ L})}{293.15 \text{ K}} = \frac{(0.03421 \text{ atm}) \cdot V_2}{223.15 \text{ K}}$$

$$V_2 = 223 \text{ L}$$

2. A 500.0 mL rigid vessel contains 0.240 mol N<sub>2</sub>, 0.450 mol H<sub>2</sub>, and 0.370 mol O<sub>2</sub> at 298 K. Calculate the partial pressure of each gas and the total pressure (in atm) of the vessel.

$$n_{\text{tot}} = n_{\text{N}_2} + n_{\text{H}_2} + n_{\text{O}_2}$$

$$= 0.240 \text{ mol} + 0.450 \text{ mol} + 0.370 \text{ mol}$$

$$n_{\text{tot}} = 1.060 \text{ mol}$$

$$P_{\text{tot}} \left( 500.0 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} \right) = (1.060 \text{ mol}) \left( 0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (298 \text{ K})$$

$$P_{\text{tot}} = 51.8 \text{ atm}$$

Partial pressure:  $P_i = \chi_i \cdot P_{\text{tot}}$ , Mole fraction:  $\chi_i = \frac{n_i}{n_{\text{tot}}}$

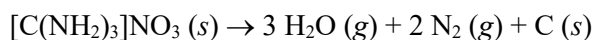
$$P_{\text{N}_2} = \frac{0.240 \text{ mol}}{1.060 \text{ mol}} \times 51.8 \text{ atm} = 11.7 \text{ atm}$$

$$P_{\text{H}_2} = \frac{0.450 \text{ mol}}{1.060 \text{ mol}} \times 51.8 \text{ atm} = 22.0 \text{ atm}$$

$$P_{\text{O}_2} = \frac{0.370 \text{ mol}}{1.060 \text{ mol}} \times 51.8 \text{ atm} = 18.1 \text{ atm}$$

$$P_{\text{tot}} = P_{\text{N}_2} + P_{\text{H}_2} + P_{\text{O}_2}$$

3. Assuming complete decomposition of 75.0 g [C(NH<sub>2</sub>)<sub>3</sub>]NO<sub>3</sub> {121.11 g/mol} at 1.00 atm and 25.0 °C, what is the total volume (in L) of gases formed by this reaction? Assume temperature and pressure are constant.

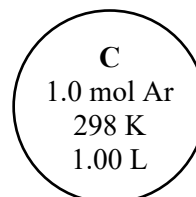
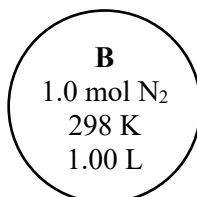
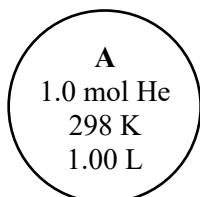


Find total number of moles of gases formed by reactant: 1.858 mol H<sub>2</sub>O (g) + 1.239 mol N<sub>2</sub> (g) = 3.096 mol

Now, calculate volume of total moles of gas.  $(1 \text{ atm}) \cdot V = (3.096 \text{ mol}) \left( 0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (298.15 \text{ K})$

$$V = 75.8 \text{ L}$$

4. Consider the following three containers (A-C) filled with gases. Assume all gases are ideal.



- (a) Which gas has the slowest root-mean-square speed ( $v_{\text{rms}}$ )?

Container C because Ar is heaviest:  $v_{\text{rms}} \propto \sqrt{\frac{1}{M}}$

- (b) Which gas exhibits the fastest rate of effusion?

Container A because He is lightest, so  $v_{\text{rms, He}} > v_{\text{rms, N}_2} > v_{\text{rms, Ar}}$

- (c) Calculate the rates of effusion for He and Ar relative to N<sub>2</sub> (i.e., by what factor are these faster or slower?).

$$\frac{v_{\text{rms, He}}}{v_{\text{rms, N}_2}} = \frac{\sqrt{28.02 \text{ g/mol}}}{\sqrt{4.00 \text{ g/mol}}} = 2.65 \text{ (FASTER)}$$

$$\frac{v_{\text{rms, Ar}}}{v_{\text{rms, N}_2}} = \frac{\sqrt{28.02 \text{ g/mol}}}{\sqrt{39.95 \text{ g/mol}}} = 0.8375 \text{ (SLOWER)}$$

## Chemical Kinetics

5. Consider the following chemical reaction:  $2 \text{NOCl}(\text{g}) \rightarrow 2 \text{NO}(\text{g}) + \text{Cl}_2(\text{g})$

If the rate of disappearance of  $\text{NOCl}(\text{g})$  is  $1.0 \times 10^{-3} \text{ M/min}$ , determine the rate of appearance of each of the two products in units of  $\text{M/min}$ .

$$-\frac{1}{2} \frac{\Delta[\text{NOCl}]}{\Delta t} = +\frac{1}{2} \frac{\Delta[\text{NO}]}{\Delta t} = +\frac{1}{1} \frac{\Delta[\text{Cl}_2]}{\Delta t}$$

NOCl is reactant, so  $\Delta[\text{NOCl}] < 0$ , extra - sign makes rate always  $> 0$ .  
 This is rate of disappearance.  
 These are rates of appearance.  
 These are stoichiometric coefficients.

$$-\frac{1}{2} \cdot (-1.0 \times 10^{-3} \frac{\text{M}}{\text{min}}) = +\frac{1}{2} \frac{\Delta[\text{NO}]}{\Delta t}$$

$$\frac{\Delta[\text{NO}]}{\Delta t} = +\frac{2}{1} \cdot \left(-\frac{1}{2}\right) \cdot (-1.0 \times 10^{-3} \frac{\text{M}}{\text{min}})$$

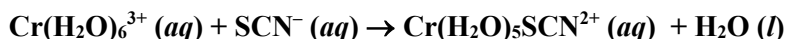
$$= +1.0 \times 10^{-3} \frac{\text{M}}{\text{min}}$$

$$-\frac{1}{2} \cdot (-1.0 \times 10^{-3} \frac{\text{M}}{\text{min}}) = +\frac{1}{1} \frac{\Delta[\text{Cl}_2]}{\Delta t}$$

$$\frac{\Delta[\text{Cl}_2]}{\Delta t} = +\frac{1}{1} \cdot \left(-\frac{1}{2}\right) \cdot (-1.0 \times 10^{-3} \frac{\text{M}}{\text{min}})$$

$$= +5.0 \times 10^{-4} \frac{\text{M}}{\text{min}}$$

6. Consider the following aqueous reaction for which initial rate data is collected at  $25^\circ\text{C}$ .



| Expt. | $[\text{Cr}(\text{H}_2\text{O})_6^{3+}]_0$ | $[\text{SCN}^-]_0$ | Initial Rate                      |
|-------|--------------------------------------------|--------------------|-----------------------------------|
| 1     | $1.0 \times 10^{-4} \text{ M}$             | 0.10 M             | $2.0 \times 10^{-11} \text{ M/s}$ |
| 2     | $1.0 \times 10^{-3} \text{ M}$             | 0.10 M             | $2.0 \times 10^{-10} \text{ M/s}$ |
| 3     | $1.5 \times 10^{-3} \text{ M}$             | 0.20 M             | $6.0 \times 10^{-10} \text{ M/s}$ |
| 4     | $1.5 \times 10^{-3} \text{ M}$             | 0.50 M             | $1.5 \times 10^{-9} \text{ M/s}$  |
| 5     | $3.0 \times 10^{-3} \text{ M}$             | 0.25 M             |                                   |

- (a) Determine the rate law for the reaction, and the value and units of the rate constant ( $k$ ).

General rate law:  $\text{Rate} = k [\text{Cr}(\text{H}_2\text{O})_6^{3+}]^a [\text{SCN}^-]^b$

Compare expts 1 and 2 to find order wrt  $\text{Cr}(\text{H}_2\text{O})_6^{3+}$ .

$$\frac{2.0 \times 10^{-10} \frac{\text{M}}{\text{s}}}{2.0 \times 10^{-11} \frac{\text{M}}{\text{s}}} = \frac{(1.0 \times 10^{-3} \text{ M})^a}{(1.0 \times 10^{-4} \text{ M})^a}$$

$$10. = (10.)^a$$

$$a = 1$$

Compare expts 3 and 4 to find order wrt  $\text{SCN}^-$ .

$$\frac{1.5 \times 10^{-9} \frac{\text{M}}{\text{s}}}{6.0 \times 10^{-10} \frac{\text{M}}{\text{s}}} = \frac{(0.50 \text{ M})^b}{(0.20 \text{ M})^b}$$

$$2.5 = (2.5)^b$$

$$b = 1$$

$$\text{Rate} = k [\text{Cr}(\text{H}_2\text{O})_6^{3+}]^1 [\text{SCN}^-]^1$$

Values from expt 1:

$$2.0 \times 10^{-11} \frac{\text{M}}{\text{s}} = k (1.0 \times 10^{-4} \text{ M})^1 (0.10 \text{ M})^1$$

$$k = \frac{2.0 \times 10^{-11} \frac{\text{M}}{\text{s}}}{(1.0 \times 10^{-4} \text{ M})^1 (0.10 \text{ M})^1}$$

$$k = 2.0 \times 10^{-6} \frac{1}{\text{M} \cdot \text{s}}$$

- (b) Determine the rate of the reaction (in  $\text{M/s}$ ) for experiment 5 in the table above.

Now, can use rate law and rate constant to predict rate at other concentrations!

$$\text{Rate} = k [\text{Cr}(\text{H}_2\text{O})_6^{3+}]^1 [\text{SCN}^-]^1$$

$$= (2.0 \times 10^{-6} \frac{1}{\text{M} \cdot \text{s}}) (3.0 \times 10^{-3} \text{ M})^1 (0.25 \text{ M})^1$$

$$= 1.5 \times 10^{-9} \frac{\text{M}}{\text{s}}$$

7. Cyclobutene rearranges (isomerizes) into butadiene according to first order kinetics with  $k = 2.0 \times 10^{-4} \text{ s}^{-1}$ .  
cyclobutene  $\rightarrow$  butadiene

In a flask, 55 torr of cyclobutene is allowed to isomerize. What will be the partial pressure, in torr, of cyclobutene after 30.0 minutes?

If first order:  $\ln[A]_t = -kt + \ln[A]_0$  OR  $\ln P_{A,t} = -kt + \ln P_{A,0}$

$$= -(2.0 \times 10^{-4} \frac{1}{s}) \left( 30.0 \text{ min} \times \frac{60 \text{ s}}{1 \text{ min}} \right) + \ln(55)$$

$$= -0.36 + 4.00$$

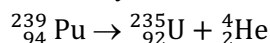
$$\ln P_{A,t} = 3.64$$

$$P_{A,t} = e^{3.64}$$

$$P_{A,t} = 38 \text{ torr}$$

**NOTE:** [A] is actually a generic term. It represents "quantity" so it's okay to work in molarity, pressure, moles, even # of atoms!

8. Plutonium-239 undergoes first-order radioactive decay with a half-life of  $^{239}\text{Pu}$  is  $t_{1/2} = 24110$  years.



Determine how long (in years) it would take for 65 % of a sample of  $^{239}\text{Pu}$  to decay.

$t_{1/2} = \frac{\ln 2}{k}$

$2.411 \times 10^4 \text{ yr} = \frac{\ln 2}{k}$

$k = 2.875 \times 10^{-5} \frac{1}{\text{yr}}$

Assume  $[\text{Pu}]_0 = 100 \text{ M}$ , so  $[\text{Pu}]_t = 35 \text{ M}$  OR  $\frac{[\text{Pu}]_t}{[\text{Pu}]_0} = 0.35$

$$\ln[A]_t = -kt + \ln[A]_0$$

$$\ln(35) = -(2.875 \times 10^{-5} \frac{1}{\text{yr}})t + \ln(100) \quad \text{OR} \quad \ln \frac{[A]_t}{[A]_0} = -kt$$

$$-1.049 = -(2.875 \times 10^{-5} \frac{1}{\text{yr}})t \quad \ln(0.35) = -(2.875 \times 10^{-5} \frac{1}{\text{yr}})t$$

$$t = 3.7 \times 10^4 \text{ yr} \quad t = 3.7 \times 10^4 \text{ yr}$$

9. Calculate  $E_a$ ,  $A$ , and  $k$  at  $499^\circ\text{C}$  using the plot and best-fit line given to the right for an unknown reaction.

slope  $= -\frac{E_a}{R}$

$$-33105 \text{ K} = -\frac{E_a}{8.314 \text{ J/mol}\cdot\text{K}}$$

$$E_a = 2.75 \times 10^5 \frac{\text{J}}{\text{mol}}$$

intercept  $= \ln(A)$

$$35.55 = \ln(A)$$

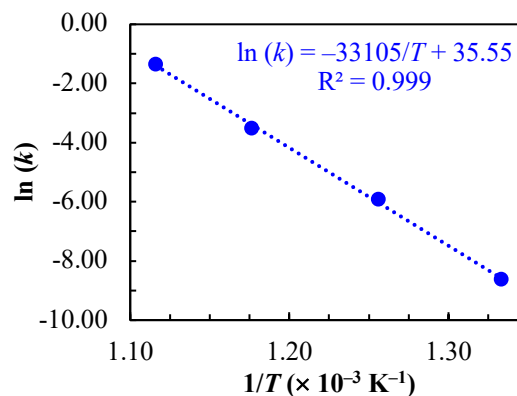
$$A = 2.75 \times 10^{15} \frac{1}{\text{s}}$$

$$k = A \cdot \exp\left(-\frac{E_a}{RT}\right)$$

$$= (2.75 \times 10^{15} \frac{1}{\text{s}}) \cdot \exp\left(-\frac{2.75 \times 10^5 \text{ J/mol}}{(8.314 \text{ J/mol}\cdot\text{K})(772.15 \text{ K})}\right)$$

$$= (2.75 \times 10^{15} \frac{1}{\text{s}}) \cdot \exp(-4.29 \times 10^1)$$

$$k = 6.6 \times 10^{-4} \frac{1}{\text{s}}$$



10. Consider the following concentration-time data for some reactant A undergoing some reaction.

| Time (s) | [A] (M) | $\ln[A]$ | $\frac{1}{[A]}$ (M <sup>-1</sup> ) |
|----------|---------|----------|------------------------------------|
| 0        | 0.240   | -1.427   | 4.17                               |
| 400      | 0.204   | -1.590   | 4.90                               |
| 800      | 0.168   | -1.784   | 5.95                               |
| 1200     | 0.132   | -2.025   | 7.58                               |
| 1600     | 0.096   | -2.343   | 10.42                              |
| 2000     | 0.060   | -2.813   | 16.67                              |

(a) Fill in the table above by calculating  $\ln[A]$  and  $\frac{1}{[A]}$  for each time given.

(b) Plot  $[A]$ ,  $\ln[A]$ , and  $\frac{1}{[A]}$  against time using the plots given to the right. Connect the data points.

(c) The order of the reaction with respect to A is ...

0<sup>th</sup> order

1<sup>st</sup> order

2<sup>nd</sup> order

(d) Determine the value and units of the rate constant ( $k$ ).

The plot of  $[A]$  vs. time is linear.

$$[A]_t = -kt + [A]_0$$

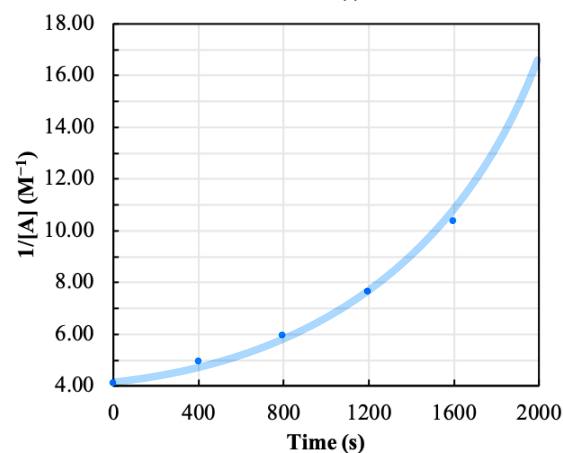
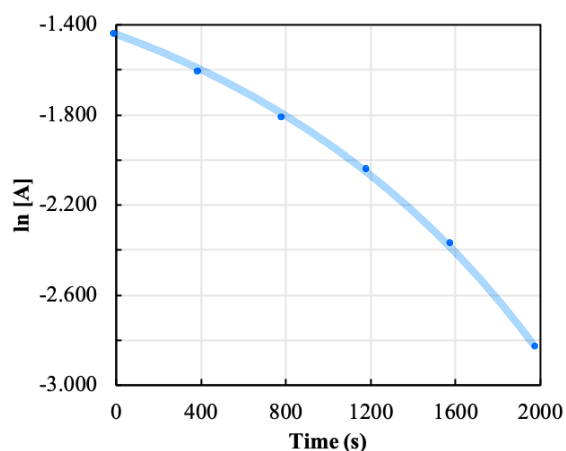
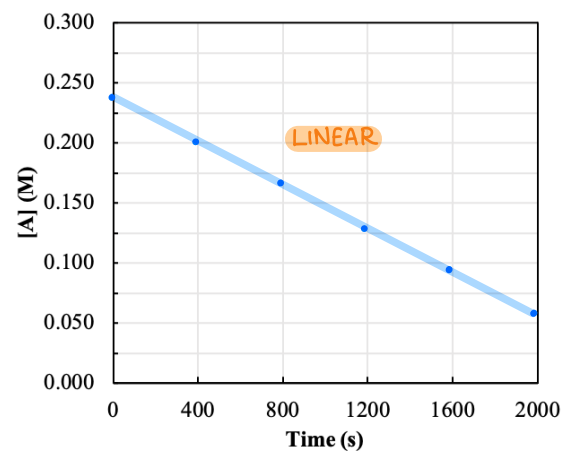
$\uparrow$     $\uparrow$     $\uparrow$     $\uparrow$   
 $y$     $=$     $m$     $x$     $+ b$

From linear plot: slope =  $m = -k \implies k = -m$

① Get slope from best-fit line or,

$$k = 9.0 \times 10^{-5} \frac{\text{M}}{\text{s}}$$

② Get slope from  $\frac{\Delta[A]}{\Delta t} = \frac{0.060 \text{ M} - 0.240 \text{ M}}{2000 \text{ s} - 0 \text{ s}} = -9.0 \times 10^{-5} \frac{\text{M}}{\text{s}}$



11. Given the proposed mechanisms for **Reaction A** below: (i) write the elementary rate law for each elementary step, (ii) write the overall balanced chemical equation, and then (iii) determine the identities of any reactants, products, intermediates, and/or catalysts in the reaction.

|                | Elementary Step                                                                         | Elementary Rate Law                                       |
|----------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------|
| step 1         | $\text{NO} + \text{NO} \rightarrow \text{N}_2\text{O}_2$                                | $\text{rate}_1 = k_1[\text{NO}]^2$                        |
| step 2         | $\text{N}_2\text{O}_2 + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{N}_2\text{O}$ | $\text{rate}_2 = k_2[\text{N}_2\text{O}_2][\text{H}_2]$   |
| step 3         | $\text{N}_2\text{O} + \text{H}_2 \rightarrow \text{N}_2 + \text{H}_2\text{O}$           | $\text{rate}_3 = k_3[\text{N}_2\text{O}][\text{H}_2]$     |
| <b>Overall</b> | $2\text{NO} + 2\text{H}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$                 |                                                           |
| Reactants:     | $\text{NO}, \text{H}_2$                                                                 | Intermediates: $\text{N}_2\text{O}_2, \text{N}_2\text{O}$ |
| Products:      | $\text{N}_2, \text{H}_2\text{O}$                                                        | Catalysts: $\text{n/a}$                                   |

formed as product in some early step,  
consumed as reactant in subsequent step(s)

12. If the observed rate law for **Reaction A** is given below, can elementary step 1 be rate-determining?

$\text{Rate} = k[\text{NO}]^2[\text{H}_2]$  ← slow elementary step for which elementary

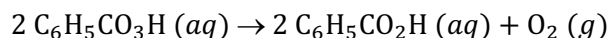
step 1:  $\text{rate}_1 = k_1[\text{NO}]^2$  → cannot be rate-determining b/c then

observed rate law would be  $\text{Rate} = k[\text{NO}]^2$

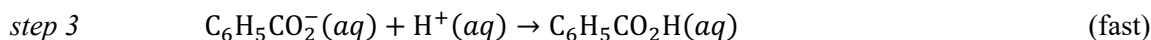
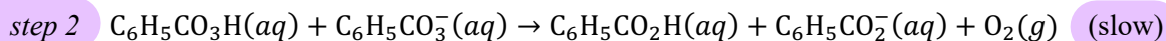
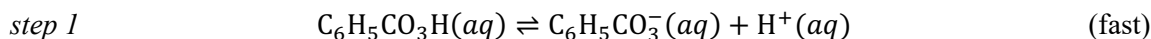
→ NOTE: This is how you would get zero order!

rate law matches observed rate law.

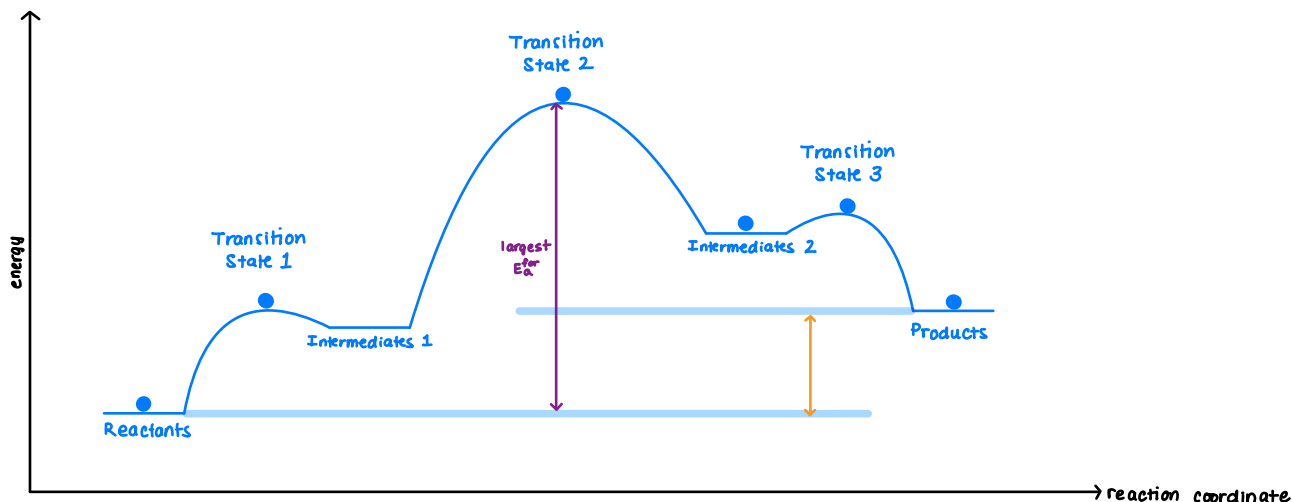
13. The overall decomposition of peroxybenzoic acid in water



is proposed to occur by the following mechanism.

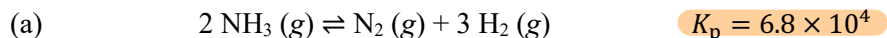


Sketch an energy diagram for the reaction and label the following: reactants, products, intermediates, transition states, and the step with the largest  $E_a$ . Assume the product state is higher in energy than the reactant state.



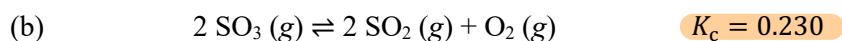
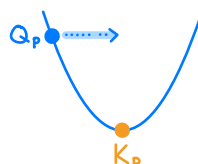
## Chemical Equilibrium

14. The chemical equations, equilibrium constants, and initial conditions are given for several reactions. For each, determine the direction in which the system will *shift* to reach equilibrium.



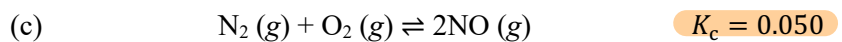
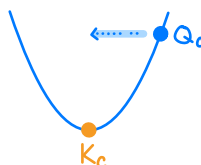
The initial partial pressures are  $P_{\text{NH}_3,0} = 2.00 \text{ atm}$ ,  $P_{\text{N}_2,0} = 10.00 \text{ atm}$ , and  $P_{\text{H}_2,0} = 10.00 \text{ atm}$ . **right**

$$Q_p = \frac{P_{\text{N}_2} P_{\text{H}_2}^3}{P_{\text{NH}_3}^2} = \frac{(10.00)(10.00)^3}{(2.00)^2} = 2.50 \times 10^3 < K_p$$



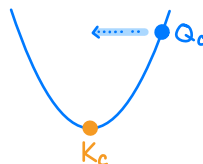
The initial concentrations are  $[\text{SO}_3]_0 = 2.00 \text{ M}$ ,  $[\text{SO}_2]_0 = 2.00 \text{ M}$ , and  $[\text{O}_2]_0 = 2.00 \text{ M}$ . **left**

$$Q_c = \frac{[\text{SO}_2]^2 [\text{O}_2]}{[\text{SO}_3]^2} = \frac{(2.00)^2 (2.00)}{(2.00)^2} = 2.00 > K_c$$

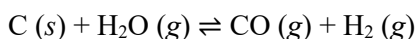


The initial concentrations are  $[\text{N}_2]_0 = 0.100 \text{ M}$ ,  $[\text{O}_2]_0 = 0.200 \text{ M}$ , and  $[\text{NO}]_0 = 1.00 \text{ M}$ . **left**

$$Q_c = \frac{[\text{NO}]^2}{[\text{N}_2] [\text{O}_2]} = \frac{(1.00)^2}{(0.100)(0.200)} = 50.0 > K_c$$



15. At elevated temperatures, carbon reacts with water vapor according to the equation:



A reaction mixture contains *only* C and  $\text{H}_2\text{O}$  initially. If  $[\text{H}_2\text{O}]_{\text{eq}} = 0.500 \text{ M}$  and  $K_c = 0.2$ , determine the equilibrium concentration of CO.

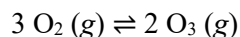
|               |   |                                                                                                                       |         |    |
|---------------|---|-----------------------------------------------------------------------------------------------------------------------|---------|----|
|               |   | $\text{C} (\text{s}) + \text{H}_2\text{O} (\text{g}) \rightleftharpoons \text{CO} (\text{g}) + \text{H}_2 (\text{g})$ |         |    |
| initial :     | I | n/a                                                                                                                   | ?       | 0M |
| change :      | C | n/a                                                                                                                   | -x      | +x |
| equilibrium : | E | n/a                                                                                                                   | 0.500 M | x  |

$$K_c = \frac{[\text{CO}][\text{H}_2]}{[\text{H}_2\text{O}]}$$

$$0.2 = \frac{x^2}{0.500}$$

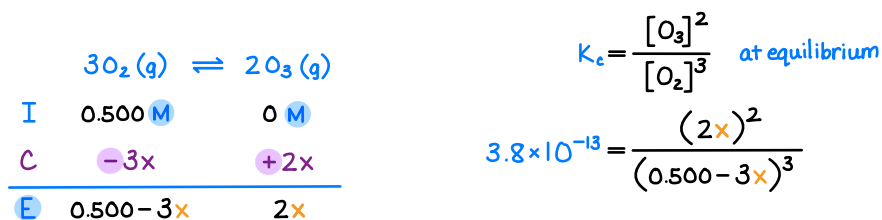
$$x = 0.3 \text{ M} \longrightarrow [\text{CO}]_{\text{eq}} = 0.3 \text{ M}$$

16. Consider the oxygen-ozone equilibrium.



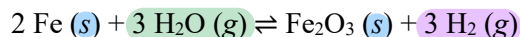
$$K_c = 3.8 \times 10^{-13} \text{ at } 2000 \text{ K}$$

If you place 1.00 mol  $\text{O}_2$  into a 2.00 L vessel at 2000 K and allow the system to reach equilibrium, what will the concentration of  $\text{O}_3$  be at equilibrium? **Set up an ICE chart and equilibrium constant expression for this problem, but do not actually solve it.**



Then solve for x and plug into equilibrium row.

17. Iron metal reacts with water vapor to produce iron(III) oxide and hydrogen gas in a vessel.



(a) Write the expression for the equilibrium constant,  $K_c$ .

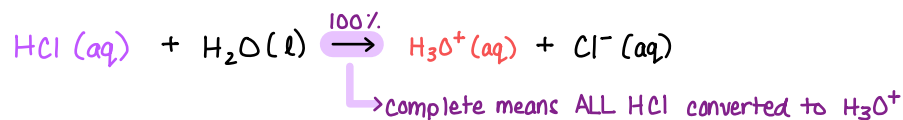
$$K_c = \frac{[\text{H}_2]^3}{[\text{H}_2\text{O}]^3}$$

(b) Suppose this system is *initially at equilibrium*. If the following changes are made to the equilibrated system, determine the direction in which the reaction will *shift/proceed* to *re-establish* equilibrium.

| Perturbation/Change/Stress |                                                                                                                         | Shift   |      |           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|---------|------|-----------|
| (i)                        | More $\text{Fe}_2\text{O}_3 (\text{s})$ is added<br><i>Solids &amp; liquids won't affect equilibrium</i>                | $Q = K$ | Left | No effect |
| (ii)                       | Some $\text{H}_2 (\text{g})$ is removed<br><i>removing product will shift rxn RIGHT to make more product</i>            | $Q < K$ | Left | No effect |
| (iii)                      | More $\text{H}_2\text{O} (\text{g})$ is added<br><i>adding reactant will shift rxn RIGHT to consume excess reactant</i> | $Q < K$ | Left | No effect |
| (iv)                       | Volume of vessel is doubled<br><i>V ↑, so P ↓, but same # moles gas, so no change</i>                                   | $Q = K$ | Left | No effect |

## Acids and Bases

18. Calculate the pH, pOH,  $[H_3O^+]$ , and  $[OH^-]$  of an aqueous solution that is 0.125 M HCl at 25 °C.

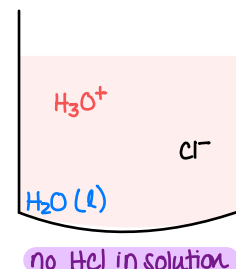


$$[H_3O^+] = \frac{0.125 \text{ mol HCl}}{1 \text{ L}} \times \frac{1 \text{ mol } H_3O^+}{1 \text{ mol HCl}} = 0.125 \text{ M } H_3O^+$$

$$pH = -\log(0.125) = 0.903$$

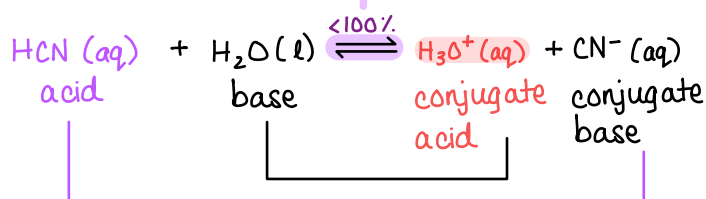
$$pOH = 14 - pH = 13.0969$$

$$[OH^-] = 10^{-pOH} = 8.00 \times 10^{-14} \quad \text{or} \quad 1 \times 10^{-14} = (0.125)[OH^-]$$



19. Consider an aqueous solution that is 0.125 M HCN at 25 °C.

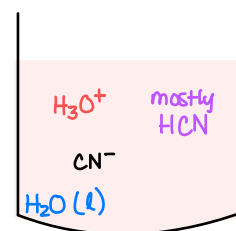
(a) Write the ionization reaction of HCN in water. equilibrium means NOT ALL HCN converted to  $H_3O^+$



(b) For the reaction above, label both sets of conjugate acid-base pairs.

(c) Write out the equilibrium constant for the reaction,  $K_a$ .

$$K_a = \frac{[H_3O^+][CN^-]}{[HCN]}$$



(d) Do you expect the pH of this solution to be greater than or less than that of 0.125 M HCl(aq)? Explain.

HCN produces less  $H_3O^+$  than HCl

because it is a weaker acid;

so  $pH(0.125 \text{ M HCN}) > pH(0.125 \text{ M HCl})$